Vaibhav Dikshit

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SUMMARY

B.Tech **Electronics and Communication Engineering** undergraduate focused on **digital hardware design, FPGA development, and verification** for embedded and networking systems. Hands-on project experience in developing RTL-based interfaces and real-time embedded solutions, followed by **simulation, testbench development, protocol validation, and hardware debugging**. Strong foundation in translating functional requirements into modular hardware designs and systematically validating corner cases and system behavior.

EDUCATION

Vellore Institute of Technology, Bhopal 2023 – 2027
B.Tech in Electronics and Communication Engineering **CGPA: 9.55/10**

KDMA International, Kanpur 2022 – 2023
Class XII (PCM) — **95%**

Gulmohar Public School, Kanpur 2020 – 2021
Class X — **95.5%**

PROJECTS

AXI4-Lite Memory-Mapped Slave IP & UVM Verification Environment GitHub
SystemVerilog, UVM, AXI4-Lite, SVA, RAL, Functional Coverage **2026**

- Implemented an **AXI4-Lite memory-mapped slave** in SystemVerilog with independent AW/W channels, byte-strobe support, register protection, and protocol-compliant response handling.
- Developed a **UVM verification environment** with Driver, Monitor, Sequencer, Scoreboard, RAL, constrained-random sequences, assertions, and regression testing.
- Verified **read/write transactions, backpressure, handshake corner cases, register accesses, and response stability** using functional coverage and SystemVerilog Assertions.

IoT-Based Vehicle Health Monitoring System GitHub
ESP32, Embedded C, Wi-Fi, MQTT, Proteus **2026**

- Designed and simulated a real-time vehicle health monitoring system integrating **3 sensors** with ESP32 and Proteus.
- Implemented threshold-based fault detection, **MQTT telemetry**, and modular Embedded C firmware with a configurable **1-second sampling interval**.

RFID-Based University ID System GitHub
ESP32, MFRC522, SPI, Python, SQLite **2025**

- Developed an ESP32-based RFID authentication system supporting **3 services**: attendance, library, and cafeteria using SPI-based UID identification.
- Built a Tkinter GUI with SQLite backend and automated Excel reporting, reducing manual record-management effort by **80%**.

Bluetooth-Controlled Smart Automation System GitHub
ESP32, Embedded C, UART, Bluetooth, Android **2025**

- Designed an ESP32-based embedded automation system enabling real-time Bluetooth communication with an Android application.
- Implemented UART-based command processing for controlling GPIO peripherals through a mobile interface.

TECHNICAL SKILLS

HDL & Verification: SystemVerilog, Verilog, UVM, SystemVerilog Assertions, RTL Design, Testbench Development
Digital & Hardware: Digital Electronics, FPGA Design, FSMs, Memories, Clocks, Resets, Microcontrollers
Protocols: AXI4-Lite, UART, SPI, I2C, CAN, Bluetooth
Embedded Systems: Embedded C, ESP32, Arduino, GPIO, Timers, Interrupts, PWM, ADC
Tools: Quartus Prime, Proteus, Keil uVision, LTSpice, Logisim, Cadence Virtuoso, GTKWave, Git, GitHub, VS Code
Programming: C, C++, Python, Java, MATLAB

CERTIFICATIONS

- **VLSI Design and Verification** — Ethnus
- **Introduction to FPGA Programming** — Univ. of Colorado
- Introduction to Programming in C — NPTEL
- Introduction to IoT — NPTEL
- MATLAB Fundamentals — MathWorks
- Fundamentals of Signal Processing — MathWorks

LEADERSHIP & EXTRACURRICULAR

- Appointed **Floor Representative**; coordinated communication between students and administration.
- Represented district in Chess; active participant in Table Tennis and Badminton.
- Organized and officiated college-level Table Tennis tournaments with **50+ participants**.